

From Platonic Solids to Regular 4-Polytopes: A Topological and Coxeter-Theoretic Classification with Lean 4 Certificate generated by LLM

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September 14, 2026

Github: <https://github.com/catskillsresearch/platonic-solids>

Abstract

We present an expository and formally verified classification of the regular convex polyhedra in $\mathbb{R}^3$ (the Platonic solids) and the regular convex polychora in $\mathbb{R}^4$, formalized in **Lean 4** against Mathlib. In dimension 3, Euler's formula $V - E + F = 2$ together with regularity incidence $pF = 2E$, $qV = 2E$ yields the Diophantine inequality $(p - 2)(q - 2) < 4$ and exactly five solutions; the same identities algebraically force $E > 0$, so non-degeneracy is a theorem rather than an extra axiom. The compared 3D theorems take $V + F = E + 2$ as a hypothesis. The development also proves algebraic Euler–Poincaré, the radial-projection homeomorphism, cellular Betti numbers of the tetrahedron surface ($\chi = 2$) and of the 4-simplex boundary ($\chi = 0$), the open-cover quasi-isomorphism in **Platonic Solids/SingularExcision/**, and singular $H_n(S^n; R) \cong R$ for $n > 0$. A chain-level bridge identifies the matrix Betti definitions with categorical homology and proves that every finite 2-complex quasi-isomorphic to singular chains of a space homotopy-equivalent to S^2 has Betti profile $(1, 0, 1)$, hence satisfies the homological Platonic classification. A fourth Palomar-compared theorem, **simplicialSingularComparison_boundaryTwo_quasiIso**, constructs the canonical realization–singular adjunction-unit chain map on the simplicial circle $\partial\Delta[2]$ and proves it is a quasi-isomorphism by comparing the simplicial Mayer–Vietoris sequence with an open-cover sequence of punctured-circle neighborhoods. This is a specific finite gluing example, not a general comparison theorem for arbitrary simplicial or polyhedral cellulations. In dimension 4, $\chi(S^3) = 0$ yields no flag bound. The compared 4D theorem says that a Platonic-cell / Platonic-vertex-figure triple with $\Delta > 0$ is one of the six regular convex 4-polytopes; the development also proves the converse and that those six are exactly the triples with $p, q, r \geq 3$ whose Gram matrix is positive definite (**regular_polychora_iff**). The classification uses the rank-4 Schläfli Gram matrix and Sylvester's criterion; it does not construct Coxeter groups as Lean objects. There is no **sorry** outside the deliberate holes in **Challenge.lean** and no project-defined axiom; the compared theorems may depend on Mathlib's **propext**, **Classical.choice**, and **Quot.sound**. Companion artifact: **Platonic Solids.lean** (standalone Lake package in this directory).

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1 Introduction: The Dimensional Hierarchy

In Euclidean geometry, a regular polytope is a figure whose symmetry group acts transitively on its flags (incident chains of vertices, edges, faces, and higher-dimensional cells).

- In **dimension 2**, there exist infinitely many regular polygons $\{p\}$ for each $p \geq 3$.
- In **dimension 3**, Euclid's *Elements* (Book XIII) establishes that there exist exactly **5 Platonic solids**.
- In **dimension 4**, Ludwig Schläfli proved in 1852 that there exist exactly **6 regular convex 4-polytopes** (polychora).
- In **dimension $n \geq 5$** , the classification collapses permanently to **3 families**: the regular simplex, the hypercube (measure polytope), and the cross-polytope.

This distribution raises foundational mathematical and pedagogical questions: **Why does dimension 3 permit 5 shapes, dimension 4 permit 6 (admitting the exceptional self-dual 24-cell), and dimension $n \geq 5$ permit only 3? Why does topology suffice to bound the classification in 3D, while failing completely in 4D?**

This paper addresses these questions through an analytical exposition coupled with a fully verified formal certificate in Lean 4. Figure 1 is the coarsest dependency blueprint: the compared 3D theorems assume Euler $V + F = E + 2$ and regularity incidence, the sign of the Schläfli determinant Δ bounds the compared 4D census, and a fourth compared theorem records the canonical simplicial-to-singular comparison for $\partial\Delta[2]$.

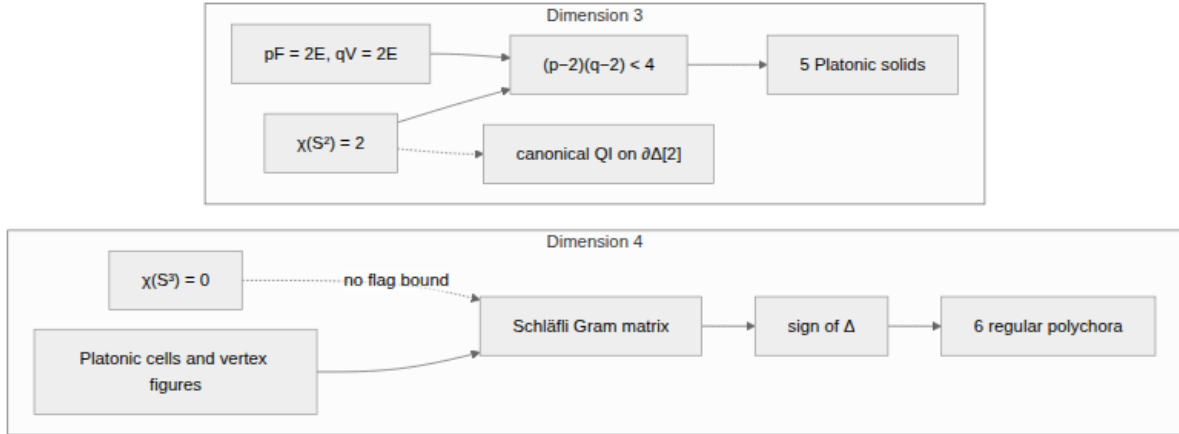


Figure 1: Coarse blueprint of the two classifications. In dimension 3, the compared theorems assume $V + F = E + 2$ and regularity incidence, which force $(p - 2)(q - 2) < 4$ and the five Platonic solids. A fourth compared theorem proves the canonical comparison on $\partial\Delta[2]$; a general cellulation comparison is not claimed. In dimension 4, $\chi(S^3) = 0$ supplies no flag bound; the compared theorem sends Platonic-compatible triples with $\Delta > 0$ to one of the six regular polychora.

2 Dimension 3: Topology, Incidence, and the Non-Degeneracy Insight

Figure 2 expands the 3D side of Figure 1 to the Lean lemmas that turn a convex polyhedron into a Platonic pair.

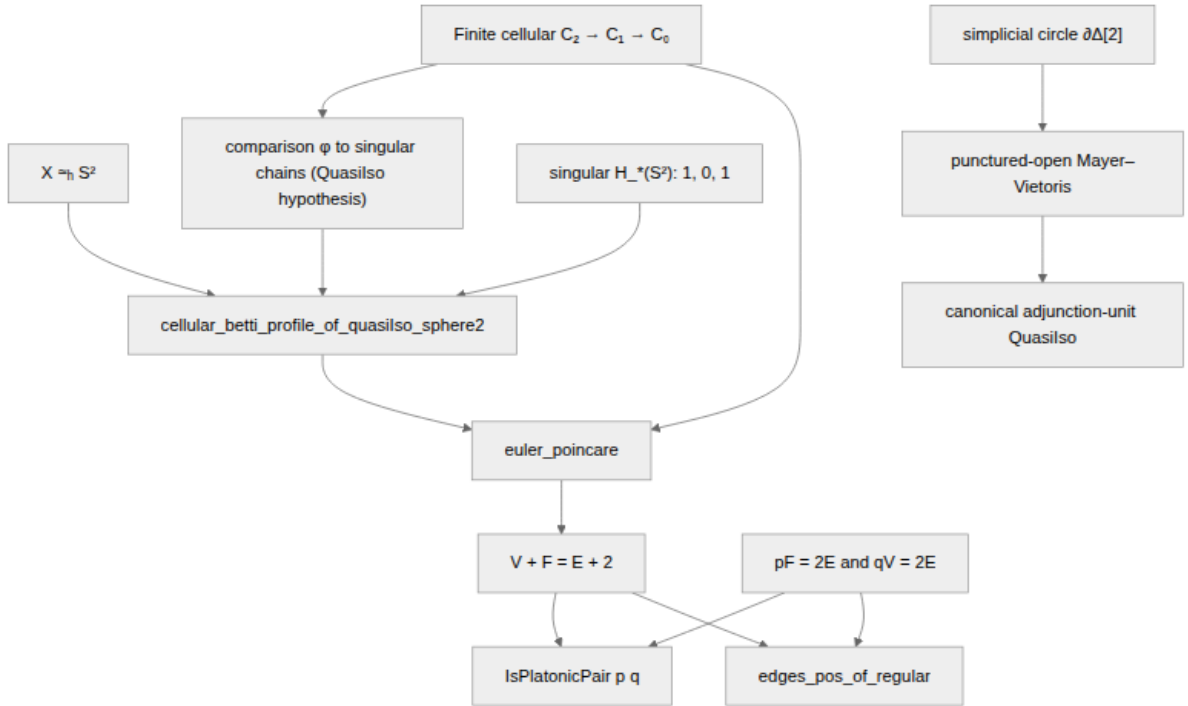


Figure 2: Dimension-3 pipeline. A quasi-isomorphism from a finite cellular complex to singular chains of a model of S^2 yields the Betti profile $(1, 0, 1)$ and hence $V + F = E + 2$. The canonical comparison is now constructed for the specific simplicial circle $\partial\Delta[2]$; arbitrary polyhedral cellulations remain separate. The compared 3D theorem continues to take Euler's formula as a hypothesis.

2.1 Topological Derivation of Euler's Formula

The Lean gists for §2.1 are the modules under `PlatonicSolids/`: radial projection and Euler–Poincaré (`RadialProjection.lean`, `EulerPoincare.lean`); cellular and singular homology (`CellularHomology.lean`, `SingularHomology.lean`, `RelativeHomology.lean`, `MayerVietoris.lean`); the excision stack (`SingularExcision/`); the simplicial–singular comparison (`SimplicialSingularComparison.lean`, `SimplicialSingularBoundaryTwo.lean`); and the sphere calculations (`SphereSingularHomology.lean`, `Sphere2Homology.lean`, `EulerBetti.lean`). Figure 5 lists every file.

Let $P \subset \mathbb{R}^3$ be a convex polyhedron containing the origin in its interior. Radial projection $\pi : \partial P \rightarrow S^2$ onto the circumscribed 2-sphere is a homeomorphism (`frontier_convex_homeo_sphere`). This projection induces a finite CW-complex structure on S^2 with V 0-cells (vertices), E 1-cells (edges), and F 2-cells (faces).

Mathlib already has the singular homology functor and its homotopy invariance. We package the missing transport lemmas: a homotopy equivalence (or homeomorphism) of spaces induces an isomorphism of singular homology (`singularHomologyIso_of_homotopyEquiv`), a contractible space has vanishing positive homology (`isZero_singularHomology_of_contractible`), and a closed disk is contractible. For the metric sphere itself, path-connectedness in dimension $n \geq 1$ gives

$$H_0(S^n; R) \cong R$$

(`singularHomology0_sphere`). The 0-sphere is the two-point set $\{\pm 1\}$, which is totally disconnected, so $H_0(S^0; R) \cong R \oplus R$ and $H_n(S^0; R) = 0$ for $n > 0$.

A subspace inclusion $A \hookrightarrow X$ is a monomorphism in `TopCat`. The singular chain-complex functor preserves monomorphisms, so

$$0 \rightarrow C_*(A; R) \rightarrow C_*(X; R) \rightarrow C_*(X, A; R) \rightarrow 0$$

is short exact (`pair_shortExact`). Snake lemma supplies the long exact sequence of the pair: connecting $\delta : H_{n+1}(X, A) \rightarrow H_n(A)$ (`pairδ`) and exactness at $H_n(A)$, $H_n(X)$, and $H_n(X, A)$ (`pair_exact_at_A`, `pair_exact_at_X`, `pair_exact_at_rel`).

For a commuting square of spaces $W \rightarrow U, V \rightarrow X$, the same argument on $C_*(W) \rightarrow C_*(U) \oplus C_*(V)$ yields the triad sequence (`triad_shortExact`, `triadδ`). The square induces a comparison from the cover chains to $C_*(X)$; the square is *excisive* when that comparison is a quasi-isomorphism (`IsExcisive`). After excision, the connecting map is Mayer–Vietoris for X (`mayerVietorisδ`).

The stereographic cover of S^n is the standard pair of charts $U = S^n \setminus \{N\}$, $V = S^n \setminus \{S\}$. Each is homeomorphic to a hyperplane (`homeoPuncturedNorth`, `homeoPuncturedSouth`), hence contractible, so $H_k(U) = H_k(V) = 0$ for $k > 0$. They cover S^n (`punctured_cover`) and meet in $S^n \setminus \{N, S\}$. Stereographic projection from the north pole sends the south pole to the origin, so $U \cap V$ is homeomorphic to the punctured hyperplane and therefore homotopy-equivalent to S^{n-1} (`homotopyEquiv_puncturedInter_sphere`).

The open-cover quasi-isomorphism is proved in `PlatonicSolids/SingularExcision/`, not imported from Mathlib. Finite ordered affine chains carry a cone-point barycentric subdivision `subdivide` and prism `prism` with $\partial \text{Sd} = \text{Sd} \partial$ and $\partial T + T \partial = \text{id} - \text{Sd}$ (`boundary_subdivide`, `boundary_prism`). Realization of an affine simplex in Δ^q is functorial on faces (`realize_face`) and on vertex-wise maps (`realize_map`), so the generators assemble to a morphism of singular chain complexes `singularSubdivision` (`ι_singularSubdivisionHom`, `singularSubdivideGenerator_d`). The prism identity is a chain homotopy $\text{id} \simeq \text{Sd}$ (`singularPrismHomotopy`). Cover chains of a two-set open cover are identified with the small-simplex subcomplex (`coverChainsIsoSmall`). Mesh shrinking is proved: each barycentric piece has diameter at most $n/(n+1)$ times its parent (`barycentricMeshRatio`, `diam_range_subdivideStd_le`), so iterated subdivision plus the Lebesgue-number lemma (`exists_subdivideStd_iterate_image_subset_left_`

or_right) puts every finite ModuleCat chain into the small-chain subgroup (eventual_small_of_chain). The algebraic deformation retract on that subgroup is instantiated (small_cycles_and_boundaries_singular). The small-subcomplex inclusion is a quasi-isomorphism (smallι_quasiIso), so coverComparison is a quasi-isomorphism and every two-set open cover is excisive (IsExcisiveSubspaces_of_openCover). The stereographic pair is therefore excisive; after the charts vanish in positive degree, Mayer–Vietoris supplies $H_{n+1}(S^{n+1}) \cong H_n(U \cap V) \cong H_n(S^n)$ for $n \geq 1$ (singularHomology_sphere_succ_iso). Degree-zero Mayer–Vietoris on S^1 identifies $H_1(S^1; R)$ with $\ker \varepsilon_{S^0} \cong R$ (singularHomology_sphere1), so $H_n(S^n; R) \cong R$ for every $n > 0$ (singularHomology_sphere) and vanishes in the other positive degrees (isZero_singularHomology_sphere).

The compared 3D classification still takes $V + F = E + 2$ as a hypothesis. The optional theorem cellular_betti_profile_of_quasiIso_sphere2 instead starts with a chain map from a finite 2-complex to the singular chains of a space $X \simeq_h S^2$ and a proof that this map is a quasi-isomorphism. CellularHomology.lean identifies categorical homology finranks with betti0, betti1, and betti2; homotopy invariance and the three singular S^2 calculations then derive the Betti profile (1, 0, 1). The corollary platonic_solids_3d_of_sphere supplies Euler–Poincaré and the incidence argument. Thus the S^2 bridge no longer assumes the three Betti numbers, although constructing that comparison from a concrete polyhedral cellulation remains an explicit input. The original three Palomar types stay combinatorial.

A fourth compared theorem records the canonical adjunction-unit comparison for the specific simplicial circle $\partial\Delta[2]$ (Palomar name: simplicialSingularComparison_boundaryTwo_quasiIso). Figure 3 records the singular-homology dependency stack.

2.1.1 The canonical comparison on the simplicial circle

For a simplicial set X , the chain map obtained by applying simplicial chains to the unit

$$X \longrightarrow \text{Sing}(|X|)$$

of the geometric-realization/singular-complex adjunction is formalized as follows (SimplicialSingularComparison.lean proves the simplex, horn $\Lambda[2, 0]$, and two-vertex cases).

```
abbrev boundaryTwoSSet : SSet.{0} :=
  (SSet.boundary 2 :
    (SSet.stdSimplex.obj [2] : SSet.{0}).Subcomplex)

abbrev singularChainsOfRealization (R : ModuleCat.{0} k) (X : SSet.{0}) :
  ChainComplex (ModuleCat.{0} k) ℕ :=
  ((singularChainComplexFunctor (ModuleCat.{0} k)).obj R).obj |X|

def simplicialSingularComparison (R : ModuleCat.{0} k) (X : SSet.{0}) :
  X.chainComplex R → singularChainsOfRealization R X :=
  SSet.chainComplexMap (sSetTopAdj.unit.app X) R
```

The boundary geometry is made concrete by the homeomorphism homeomorph_boundaryTwo_addCircle,

$$|\partial\Delta[2]| \simeq \text{AddCircle}(3).$$

The obvious decomposition into one closed edge and the complementary two-edge horn gives a short exact sequence on simplicial chains, but the analogous sequence of all singular chains on the two closed arcs is not short exact: a singular simplex in the circle need not lie wholly in either closed arc.

SimplicialSingularBoundaryTwo.lean enlarges the arcs to two punctured-circle open neighborhoods. Each neighborhood deformation retracts to its edge or horn, while their intersection deformation retracts to the two shared vertices. These three equivalences give quasi-isomorphisms on singular chains. The source simplicial Mayer–Vietoris sequence is then mapped to the target

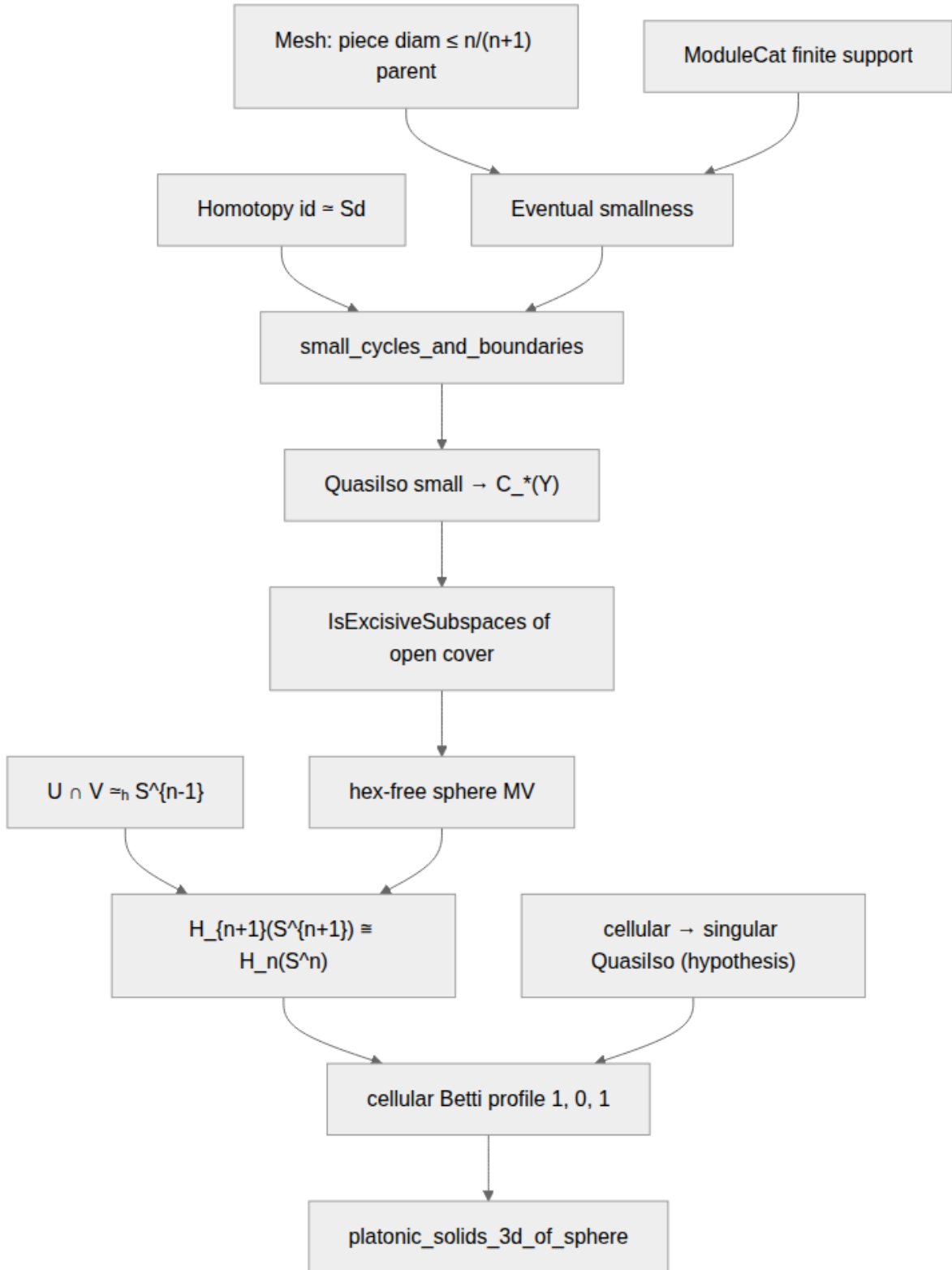


Figure 3: Singular-excision blueprint, finer than Figure 2. Mesh shrinking and the prism homotopy feed the small-chain retract; the open-cover quasi-isomorphism discharges **IsExcisiveSubspaces** and yields $H_{n+1}(S^{n+1}) \cong H_n(S^n)$.

open-cover short exact sequence; two-out-of-three proves the induced map to cover chains is a quasi-isomorphism, and open-cover excision identifies cover chains with the singular chains of the whole circle. The resulting compared theorem is as follows.

Compared theorem (`simplicialSingularComparison_boundaryTwo_quasiIso`). For any coefficient module R , the chain map induced by the realization–singular adjunction unit

$$C_*^{\text{simp}}(\partial\Delta[2]; R) \longrightarrow C_*^{\text{sing}}(|\partial\Delta[2]|; R)$$

is a quasi-isomorphism.

```
theorem simplicialSingularComparison_boundaryTwo_quasiIso
  (R : ModuleCat.{0} k) :
  QuasiIso (simplicialSingularComparison R boundaryTwoSSet) := by
  ...
```

The normalized-chain corollary is supporting material and is not compared. No theorem for arbitrary finite nonsingular simplicial sets or arbitrary polyhedral cellulations is claimed.

The standard simplicial 2-sphere is the surface of a tetrahedron: 4 vertices, 6 edges, 4 triangles. Oriented incidence matrices $d_1 : \mathbb{Q}^6 \rightarrow \mathbb{Q}^4$ and $d_2 : \mathbb{Q}^4 \rightarrow \mathbb{Q}^6$ satisfy $d_1 d_2 = 0$. Each has a 3×3 identity minor, so both ranks are at least 3; the product-zero rank inequality on a 6-dimensional middle term forces both ranks equal to 3. Hence

$$b_0 = 4 - 3 = 1, \quad b_1 = 3 - 3 = 0, \quad b_2 = 4 - 3 = 1$$

(`homology_sphere2`). The Euler characteristic is the alternating sum of Betti numbers:

$$\chi(S^2) = b_0 - b_1 + b_2 = 1 - 0 + 1 = 2$$

Algebraic Euler–Poincaré (`euler_poincare`) is rank-nullity on any 2-complex: cell ranks equal Betti ranks. Specializing to these numbers yields Euler’s formula

$$V - E + F = 2$$

on the tetrahedron (`euler_tetrahedron`) and, for an arbitrary 2-complex with the same Betti numbers, the general identity `euler_formula_of_sphere2`. The tetrahedron incidence structure is then Platonic (`tetrahedron_platonic`). Combining Euler–Poincaré with regularity incidence gives the classification from homology (`platonic_solids_3d_of_homology`).

The two routes are now joined at the correct level of abstraction. Given a quasi-isomorphism from a finite cellular chain complex to singular chains of a space homotopy-equivalent to S^2 , the Betti profile is forced and the Platonic classification follows.

```
theorem cellular_betti_profile_of_quasiIso_sphere2
  ... (φ : cellularChainComplex ... → singular chains of X) [QuasiIso φ]
  (e : X ≃_h MetricSphere 2) :
  betti0 d_1 = 1 ∧ betti1 d_2 d_1 = 0 ∧ betti2 d_2 = 1 := by
  ...
```

```
theorem platonic_solids_3d_of_sphere
  ... [QuasiIso φ] (e : X ≃_h MetricSphere 2) :
  IsPlatonicPair p q := by
  ...
```

The canonical comparison has been constructed for $\partial\Delta[2]$ as a first finite gluing case; extending it naturally to general finite simplicial cellulations, and then to the polyhedral cellulations needed here, remains open.

2.2 Incidence Combinatorics

The Lean gist is `PlatonicSolids/Incidence.lean` (with the pair predicate in `PlatonicSolids/PlatonicPair.lean`). A regular polyhedron has Schläfli symbol $\{p, q\}$, meaning: 1. Every 2D face is an identical regular p -gon ($p \geq 3$). Each face contributes p edge-face incidences. Since every edge is shared by exactly 2 faces:

$$pF = 2E \implies F = \frac{2E}{p}$$

2. Every vertex meets q faces and q edges ($q \geq 3$). Each vertex contributes q edge-vertex incidences. Since every edge connects exactly 2 vertices:

$$qV = 2E \implies V = \frac{2E}{q}$$

2.3 The Diophantine Reduction

The Lean gist is `PlatonicSolids/PlatonicPair.lean`; the reciprocal form $1/p + 1/q > 1/2$ is `PlatonicSolids/Reciprocal.lean`. Substituting these incidence relations into Euler's formula:

$$\frac{2E}{q} - E + \frac{2E}{p} = 2$$

Multiplying by pq over $\mathbb{N}$ avoids rational arithmetic:

$$(2p + 2q)E = (pq)E + 2pq$$

Rearranging terms:

$$(2p + 2q - pq)E = 2pq$$

Because $2pq > 0$ for $p, q \geq 3$, the factor $(2p + 2q - pq)$ must be strictly positive, which yields:

$$pq < 2p + 2q \iff (p - 2)(q - 2) < 4$$

Since $p \geq 3$ and $q \geq 3$, both $(p - 2)$ and $(q - 2)$ are strictly positive integers. The only pairs of positive integers with product strictly less than 4 are:

$(p - 2)$	$(q - 2)$	$(p - 2)(q - 2)$	Pair (p, q)	Polyhedron
1	1	1	(3, 3)	Tetrahedron
1	2	2	(3, 4)	Octahedron
2	1	2	(4, 3)	Cube (Hexahedron)
1	3	3	(3, 5)	Icosahedron
3	1	3	(5, 3)	Dodecahedron

The five pairs are bundled as a predicate on indices:

```
def IsPlatonicPair (p q : ℕ) : Prop :=
  (p = 3 ∧ q = 3) ∨ (p = 3 ∧ q = 4) ∨ (p = 4 ∧ q = 3) ∨
  (p = 3 ∧ q = 5) ∨ (p = 5 ∧ q = 3)
```

Incidence and Euler then yield the compared 3D classification:

```
theorem platonic_solids_3d
  (V E F p q : ℕ) (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E) (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  IsPlatonicPair p q := by
  ...
```

2.4 Formalization Insight: Elimination of the Non-Degeneracy Axiom

The Lean gist is `edges_pos_of_regular` in `PlatonicSolids/Incidence.lean`. In informal presentations, dividing Euler’s equation by $2E$ to obtain:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{2} + \frac{1}{E} > \frac{1}{2}$$

is typically justified by appealing to physical or geometric intuition: “a solid polyhedron clearly has edges, so $E > 0$.”

When formalizing this theorem in Lean 4, one might initially expect to require an explicit non-degeneracy hypothesis (`hE : E > 0`). However, machine verification reveals this hypothesis to be mathematically redundant.

Recall the cleared identity over $\mathbb{N}$:

$$(2p + 2q)E = (pq)E + 2pq$$

If $E = 0$, the equation reduces to:

$$0 = 2pq$$

Because $p \geq 3$ and $q \geq 3$, we have $2pq \geq 18$, which contradicts $0 = 2pq$.

Therefore, **the topological Euler characteristic ($\chi = 2$) together with the local regularity conditions ($p, q \geq 3$) algebraically prohibits an edgeless configuration**. The condition $E \geq 1$ is an algebraic consequence of the system, not an extrinsic geometric requirement.

```
theorem edges_pos_of_regular
  (V E F p q : ℕ) (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E) (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  0 < E := by
  ...
```

3 Dimension 4: Why Topology Fails and the Gram Matrix Takes Over

Figure 4 is the 4D counterpart of Figure 2: Platonic cells and vertex figures determine a Schläfli Gram matrix, and the sign of $\Delta = \det M$ sorts the eleven triples.

3.1 The Topological Barrier: $\chi(S^3) = 0$

The Lean gists are `PlatonicSolids/EulerPoincare.lean` (now also the 3-complex identity `euler_poincare3`), `PlatonicSolids/Sphere3Homology.lean` (cellular $H_*(S^3; \mathbb{Q})$ of the 4-simplex boundary), and `eulerChar_sphere3_betti` in `PlatonicSolids/EulerBetti.lean`.

A convex 4-polytope with nonempty interior has frontier homeomorphic to S^3 by the same radial projection as in §2.1. The standard simplicial 3-sphere is the boundary of a 4-simplex: 5 vertices, 10 edges, 10 triangles, 5 tetrahedra. Oriented incidence matrices $d_1 : \mathbb{Q}^{10} \rightarrow \mathbb{Q}^5$, $d_2 : \mathbb{Q}^{10} \rightarrow \mathbb{Q}^{10}$, $d_3 : \mathbb{Q}^5 \rightarrow \mathbb{Q}^{10}$ satisfy $d_1 d_2 = 0$ and $d_2 d_3 = 0$. Identity minors give $\text{rank } d_1 \geq 4$ and $\text{rank } d_3 \geq 4$; a 6×6 minor of d_2 has determinant 1, so $\text{rank } d_2 \geq 6$. The product-zero inequalities on the two middle dimensions of size 10 then force the ranks to be exactly 4, 6, and 4. Hence

$$b_0 = 5 - 4 = 1, \quad b_1 = (10 - 4) - 6 = 0, \quad b_2 = (10 - 6) - 4 = 0, \quad b_3 = 5 - 4 = 1$$

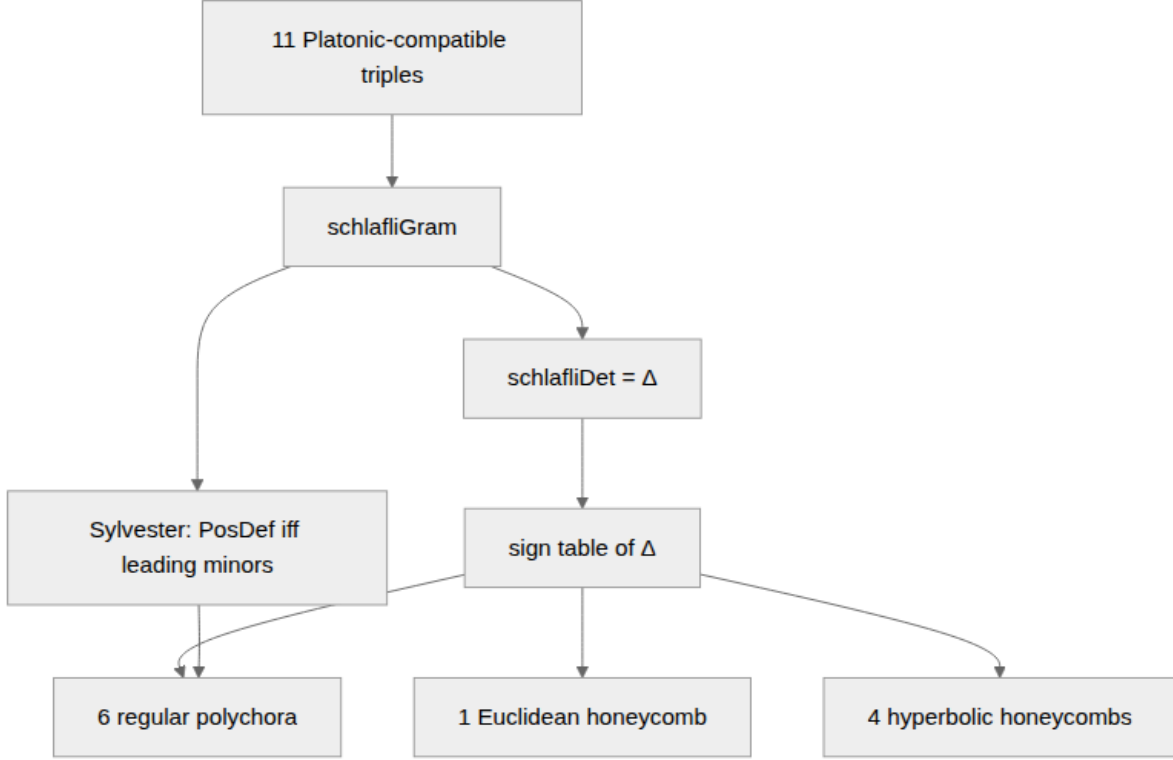


Figure 4: Dimension-4 pipeline. The eleven Platonic-compatible Schläfli triples are classified by the sign of $\Delta = \det(\text{schlaflGram})$. Positive Δ is equivalent, by Sylvester, to positive-definiteness of the Gram matrix, and picks out the six regular convex polychora.

(`homology_sphere3`). The Euler characteristic vanishes:

$$\chi(S^3) = b_0 - b_1 + b_2 - b_3 = 1 - 0 + 0 - 1 = 0$$

Algebraic Euler–Poincaré in dimension 3 (`euler_poincare3`) again equates cell ranks with Betti ranks, so any such cellulation satisfies

$$V - E + F - C = 0$$

(`eulerChar_sphere3` on the 4-simplex, and `euler_formula_of_sphere3` in general). Because the alternating sum is zero, dividing by any flag count yields an identity equal to zero rather than a positive reciprocal like $2/E$. **Topology alone does not impose an upper bound on p, q , or r in 4D.**

3.2 Coxeter Systems and the Schläfli Gram Matrix

The Lean gist is `PlatonicSolids/GramMatrix.lean`. To obtain an upper bound, informal accounts invoke **Coxeter reflection groups**. The Lean development does not construct those groups; it works with the associated **Schläfli Gram matrix** and its determinant Δ . A regular 4-polytope is described by its Schläfli symbol $\{p, q, r\}$: * The 3D cells are Platonic solids $\{p, q\}$. * The 3D vertex figures are Platonic solids $\{q, r\}$. * r cells meet around every 1D edge.

The Coxeter group W associated with $\{p, q, r\}$ is generated by reflections $\{s_1, s_2, s_3, s_4\}$ across the facet hyperplanes of a spherical fundamental simplex:

$$W = \langle s_1, s_2, s_3, s_4 \mid (s_i s_j)^{m_{ij}} = 1 \rangle$$

where the exponents are $m_{ii} = 1$, $m_{12} = p$, $m_{23} = q$, $m_{34} = r$, and $m_{ij} = 2$ for $|i - j| > 1$.

The canonical bilinear form of this system is represented by the **Schläfli Gram matrix**:

$$M = \begin{pmatrix} 1 & -\cos \frac{\pi}{p} & 0 & 0 \\ -\cos \frac{\pi}{p} & 1 & -\cos \frac{\pi}{q} & 0 \\ 0 & -\cos \frac{\pi}{q} & 1 & -\cos \frac{\pi}{r} \\ 0 & 0 & -\cos \frac{\pi}{r} & 1 \end{pmatrix}$$

In Lean the matrix and scalar determinant are named as follows:

```
noncomputable def schlafligram (p q r : ℕ) : Matrix (Fin 4) (Fin 4) ℝ :=
  !![1, -cos (π / p), 0, 0;
    -cos (π / p), 1, -cos (π / q), 0;
    0, -cos (π / q), 1, -cos (π / r);
    0, 0, -cos (π / r), 1]

noncomputable def schlaflidet (p q r : ℕ) : ℝ :=
  sin (π / p) ^ 2 * sin (π / r) ^ 2 - cos (π / q) ^ 2
```

3.3 Sylvester's Criterion and Principal Minors

The Lean gists are `PlatonicSolids/Angles.lean`, `LeadingMinors.lean`, `TrailingMinor.lean`, `D3IfPlatonic.lean`, `QuadForm.lean`, `CompletingSquare.lean`, and `Sylvester.lean`. By the Cartan–Killing–Coxeter classification: 1. M is **positive definite** ($\det(M) > 0$) $\iff W$ is a **finite reflection group acting on S^3 (a finite regular convex 4-polytope)**. 2. M is **positive semidefinite with nullity 1** ($\det(M) = 0$) $\iff W$ is an **affine reflection group (a Euclidean 3D space-filling honeycomb)**. 3. M has **negative eigenvalues** ($\det(M) < 0$) $\iff W$ is a **hyperbolic reflection group (a honeycomb in hyperbolic space $\mathbb{H}^3$)**.

By Sylvester's criterion, M is positive definite if and only if all leading principal minors $D_k = \det(M_k)$ are strictly positive: * $D_1 = 1 > 0$. * $D_2 = 1 - \cos^2 \frac{\pi}{p} = \sin^2 \frac{\pi}{p} > 0$ (satisfied for all $p \geq 3$). * $D_3 = \sin^2 \frac{\pi}{p} - \cos^2 \frac{\pi}{q}$.

$$D_3 > 0 \iff \sin \frac{\pi}{p} > \cos \frac{\pi}{q} = \sin \left(\frac{\pi}{2} - \frac{\pi}{q} \right) \iff \frac{1}{p} + \frac{1}{q} > \frac{1}{2}$$

Thus, $D_3 > 0$ requires the cell $\{p, q\}$ to be a Platonic solid. * Symmetrically, the trailing 3×3 minor requires:

$$\frac{1}{q} + \frac{1}{r} > \frac{1}{2}$$

Thus, the vertex figure $\{q, r\}$ must also be a Platonic solid.

3.4 The Critical Dimension-4 Minor: D_4

The Lean gist is `PlatonicSolids/DetFormula.lean`. Expanding $D_4 = \det(M)$ along the fourth row:

$$D_4 = D_3 - \cos^2 \left(\frac{\pi}{r} \right) D_2 = \left(\sin^2 \frac{\pi}{p} - \cos^2 \frac{\pi}{q} \right) - \cos^2 \left(\frac{\pi}{r} \right) \sin^2 \left(\frac{\pi}{p} \right)$$

Factoring $\sin^2 \frac{\pi}{p}$:

$$D_4 = \sin^2 \left(\frac{\pi}{p} \right) \left(1 - \cos^2 \frac{\pi}{r} \right) - \cos^2 \left(\frac{\pi}{q} \right) = \sin^2 \left(\frac{\pi}{p} \right) \sin^2 \left(\frac{\pi}{r} \right) - \cos^2 \left(\frac{\pi}{q} \right)$$

Therefore, the condition that $\{p, q, r\}$ forms a finite regular convex 4-polytope is:

$$\Delta(p, q, r) := \sin^2 \left(\frac{\pi}{p} \right) \sin^2 \left(\frac{\pi}{r} \right) - \cos^2 \left(\frac{\pi}{q} \right) > 0$$

4 Evaluation of the 11 Candidate Triples

The Lean gists are `CompatibleTriples.lean`, `TrigValues.lean`, `SignTable.lean`, `Regular4.lean`, and `Classification.lean` in `PlatonicSolids/`.

Because both $\{p, q\}$ and $\{q, r\}$ must be Platonic, the indices must satisfy $p, q, r \in \{3, 4, 5\}$. Among the $5 \times 5 = 25$ combinations, only **11 triples** have a compatible middle index q .

The exact squared trigonometric values are:

$$\begin{aligned}\cos^2 \frac{\pi}{3} &= \frac{1}{4}, & \sin^2 \frac{\pi}{3} &= \frac{3}{4} \\ \cos^2 \frac{\pi}{4} &= \frac{1}{2}, & \sin^2 \frac{\pi}{4} &= \frac{1}{2} \\ \cos^2 \frac{\pi}{5} &= \frac{3 + \sqrt{5}}{8}, & \sin^2 \frac{\pi}{5} &= \frac{5 - \sqrt{5}}{8}\end{aligned}$$

Evaluating $\Delta(p, q, r)$ across all 11 candidate triples:

4.1 Group I: Spherical Polychora ($\Delta > 0$)

1. $\{3, 3, 3\}$ (**5-cell / Simplex**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{1}{4} = \frac{9}{16} - \frac{4}{16} = \frac{5}{16} > 0$$

2. $\{3, 3, 4\}$ (**16-cell**) & $\{4, 3, 3\}$ (**8-cell / Tesseract**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{1}{2}\right) - \frac{1}{4} = \frac{3}{8} - \frac{2}{8} = \frac{1}{8} > 0$$

3. $\{3, 4, 3\}$ (**24-cell — Exceptional Self-Dual Polytope**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{1}{2} = \frac{9}{16} - \frac{8}{16} = \frac{1}{16} > 0$$

4. $\{3, 3, 5\}$ (**600-cell**) & $\{5, 3, 3\}$ (**120-cell**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{5 - \sqrt{5}}{8}\right) - \frac{1}{4} = \frac{7 - 3\sqrt{5}}{32}$$

Since $7^2 = 49 > (3\sqrt{5})^2 = 45$, we have $7 - 3\sqrt{5} > 0$, so $\Delta > 0$.

4.2 Group II: Euclidean Honeycomb ($\Delta = 0$)

5. $\{4, 3, 4\}$ (**Cubic Honeycomb of $\mathbb{R}^3$**):

$$\Delta = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) - \frac{1}{4} = \frac{1}{4} - \frac{1}{4} = 0 \quad (\text{Affine / flat space-filling})$$

4.3 Group III: Hyperbolic Honeycombs ($\Delta < 0$)

6. $\{3, 5, 3\}$ (Icosacell Honeycomb):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{3 + \sqrt{5}}{8} = \frac{3 - 2\sqrt{5}}{16} < 0 \quad (\text{since } 3^2 = 9 < 20)$$

7. $\{4, 3, 5\}$ & $\{5, 3, 4\}$:

$$\Delta = \left(\frac{1}{2}\right) \left(\frac{5 - \sqrt{5}}{8}\right) - \frac{1}{4} = \frac{1 - \sqrt{5}}{16} < 0$$

8. $\{5, 3, 5\}$:

$$\Delta = \left(\frac{5 - \sqrt{5}}{8}\right)^2 - \frac{1}{4} = \frac{14 - 10\sqrt{5}}{64} < 0 \quad (\text{since } 14^2 = 196 < 500)$$

Thus, **precisely 6 triples yield positive-definite Gram matrices.**

The six spherical triples are packaged as `IsRegular4Polytope`; the compared 4D theorem is:

```
def IsRegular4Polytope (p q r : ℕ) : Prop :=
  (p = 3 ∧ q = 3 ∧ r = 3) ∨ (p = 4 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 4) ∨ (p = 3 ∧ q = 4 ∧ r = 3) ∨
  (p = 5 ∧ q = 3 ∧ r = 3) ∨ (p = 3 ∧ q = 3 ∧ r = 5)

theorem regular_polychora_classification (p q r : ℕ)
  (h_cell : IsPlatonicPair p q) (h_vf : IsPlatonicPair q r)
  (h_det : 0 < schlaflidDet p q r) :
  IsRegular4Polytope p q r := by
  ...
```

5 Complete Lean 4 Formal Verification

The Lake package compiles against Lean 4 and Mathlib. It contains no project-defined axiom, no `admit`, and no `sorry` outside the deliberate holes in `Challenge.lean`. Compared theorems may still depend on Mathlib’s `propext`, `Classical.choice`, and `Quot.sound`. The development is split into section-sized modules under `PlatonicSolids/`; gists in the main text quote theorem headers and definitions, and the appendix indexes every module with a hyperlink to [the GitHub repository](#). `PlatonicSolids.lean` is the root import. Figures 5 and 6 are the module-level blueprints: first the top-level `PlatonicSolids/*.lean` graph, then the `SingularExcision/` subgraph.

The compared theorems are (Palomar statement of record in `Challenge.lean`; proofs in the modules cited above):

- `platonic_solids_3d` — Euler $V + F = E + 2$ (as a hypothesis) plus regularity incidence $pF = 2E$, $qV = 2E$ force a Platonic pair (p, q) .

```
theorem platonic_solids_3d
  (V E F p q : ℕ) (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E) (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  IsPlatonicPair p q := by sorry
```


- `edges_pos_of_regular` — the same identities algebraically force $E > 0$; non-degeneracy is a theorem, not an axiom.

```
theorem edges_pos_of_regular
  (V E F p q : ℕ) (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E) (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  0 < E := by sorry
```

- `regular_polychora_classification` — Platonic cells and vertex figures with $\Delta(p, q, r) > 0$ force one of the six regular convex 4-polytopes.

```
theorem regular_polychora_classification (p q r : ℕ)
  (h_cell : IsPlatonicPair p q) (h_vf : IsPlatonicPair q r)
  (h_det : 0 < schlafliDet p q r) :
  IsRegular4Polytope p q r := by sorry
```

- `simplicialSingularComparison_boundaryTwo_quasiIso` — the canonical adjunction-unit chain map on $\partial\Delta[2]$ is a quasi-isomorphism.

```
theorem simplicialSingularComparison_boundaryTwo_quasiIso
  (R : ModuleCat.{0} k) :
  QuasiIso (simplicialSingularComparison R boundaryTwoSSet) := by sorry
```

The converse $\Delta > 0$ for each of the six is `schlafliDet_pos_of_regular`. The stronger equivalence `regular_polychora_iff` identifies those six with positive-definiteness of `schlafliGram` for $p, q, r \geq 3$, via Sylvester’s criterion (`PlatonicSolids/Sylvester.lean`). Neither is a Palomar compared declaration.

`Challenge.lean` restates the compared family with deliberate `sorry`s so Palomar Comparator can check types against `Solution.lean`. Supporting definitions include `IsPlatonicPair`, `IsRegular4Polytope`, `schlafliGram`, and `schlafliDet`, together with the boundary-comparison helpers `boundaryTwoSSet`, `singularChainsOfRealization`, and `simplicialSingularComparison`.

File	Paper	Content
<code>PlatonicSolids/RadialProjection.lean</code>	§2.1	Frontier of a convex body $\simeq$ sphere
<code>PlatonicSolids/EulerPoincare.lean</code>	§§2.1, 3.1	Algebraic Euler–Poincaré in dimensions 2 and 3
<code>PlatonicSolids/CellularHomology.lean</code>	§2.1	Categorical finite 2-complex; Betti finranks; quasi-isomorphism transport
<code>PlatonicSolids/SingularHomology.lean</code>	§2.1	Homotopy invariance; $H_0(S^n)$; $H_*(S^0)$
<code>PlatonicSolids/RelativeHomology.lean</code>	§2.1	Pair sequence $0 \rightarrow C_*(A) \rightarrow C_*(X) \rightarrow C_*(X, A) \rightarrow 0$ and LES
<code>PlatonicSolids/MayerVietoris.lean</code>	§2.1	Triad SES, excision predicate, stereographic cover of S^n
<code>PlatonicSolids/SingularExcision.lean</code>	§2.1	Root of the barycentric / small-chain stack

File	Paper	Content
PlatonicSolids/Singular Excision/*.lean	§2.1	Mesh shrinking, prism homotopy, small retract, cover = small, open-cover QI
SimplicialSingular Comparison.lean	§2.1	Adjunction-unit comparison; simplex, horn, two-vertex cases
SimplicialSingular BoundaryTwo.lean	§2.1	Open-cover proof for $\partial\Delta[2]$
SphereSingularHomology. lean	§2.1	$H_n(S^n; R) \cong R$; cellular S^2 bridge
Sphere2Homology.lean	§2.1	Cellular $H_*(S^2; \mathbb{Q})$ of tetrahedron surface
Sphere3Homology.lean	§3.1	Cellular $H_*(S^3; \mathbb{Q})$ of 4-simplex boundary
PlatonicSolids/Euler Betti.lean	§§2.1, 3.1	$\chi(S^2) = 2$, $\chi(S^3) = 0$
PlatonicSolids/Platonic Pair.lean	§2.3	Five solutions of $(p-2)(q-2) < 4$
PlatonicSolids/ Reciprocal.lean	§§2.3, 3.3	$1/p + 1/q > 1/2$
PlatonicSolids/Incidence. lean	§§2.2–2.4	platonic_solids_3d, edges_ pos_of_regular
PlatonicSolids/Angles. lean	§3.3	$\pi/n \in (0, \pi/2)$
PlatonicSolids/Gram Matrix.lean	§3.2	schlafligram, schlafliDet
PlatonicSolids/Leading Minors.lean	§3.3	D_1, D_2, D_3
PlatonicSolids/Trailing Minor.lean	§3.3	Vertex-figure 3×3
PlatonicSolids/D3Iff Platonic.lean	§3.3	$D_3 > 0$ iff Platonic cell
PlatonicSolids/QuadForm. lean	§3.3	Gram quadratic form
PlatonicSolids/Completing Square.lean	§3.3	LDL identity
PlatonicSolids/Det Formula.lean	§3.4	$\det M = \Delta$
PlatonicSolids/Sylvester. lean	§3.3	PosDef iff leading minors > 0
PlatonicSolids/Trig Values.lean	§4	$\sin^2(\pi/n)$, $\cos^2(\pi/n)$ for $n \in \{3, 4, 5\}$
PlatonicSolids/Compatible Triples.lean	§4	Eleven compatible triples
PlatonicSolids/SignTable. lean	§4	Sign of Δ
PlatonicSolids/Regular4. lean	§4	The six regular 4-polytopes

The remaining named lemmas are the computational steps of those paragraphs: `platonic_pairs_of_inequality` / `platonic_pair_iff` / `platonic_pair_iff_reciprocal` for the Diodorine pair; `leadingMinor_one/two/three` and `leadingMinor_three_pos_iff` / `trailing3_pos_iff` for Sylvester’s first three minors; `schlafligram_quad` / `band_quad_squares` / `schlafligram_det` for $\det M = \Delta$; `schlafliDet_*` and `schlafliDet_*_pos/neg` for the eleven signs; `regular_polychora_iff` / `posDef_of_regular` / `regular_of_posDef` for the PosDef equivalence.

6 Pedagogical and Formalization Significance

1. **Foundational Consistency Across Dimensions:** The mathematical transition between 3D and 4D is made explicit: the compared 3D theorems take Euler $V + F = E + 2$ as a hypothesis, while the chain-level theorem derives it whenever the cellular complex is quasi-isomorphic to singular chains of an S^2 model. The fourth compared result verifies the canonical comparison in the first noncontractible finite simplicial gluing case, $\partial\Delta[2]$, without presenting it as the still-open general cellulation theorem. The compared 4D theorem is instead bounded by the sign of Δ ; Sylvester’s criterion identifies the six with positive-definiteness of the Gram matrix in the uncomparing `regular_polychora_iff`.
2. **Elimination of Redundant Hypotheses:** As demonstrated in `edges_pos_of_regular`, Lean’s linter and tactic engine show that the geometric non-degeneracy condition $E > 0$ is a theorem rather than an independent axiom.
3. **Absence of Ad-Hoc Combinatorial Filters:** Candidate 4D triples are filtered using the **Schläfli determinant** $\Delta(p, q, r) = \sin^2 \frac{\pi}{p} \sin^2 \frac{\pi}{r} - \cos^2 \frac{\pi}{q}$. Every rejection is certified by non-linear arithmetic over real algebraic numbers, with all properties of $\sqrt{5}$ derived from first principles.
4. **An explicit finite comparison, not a general one:** Closed-arc singular chains on $|\partial\Delta[2]|$ do not form a short exact sequence. The compared proof therefore enlarges the last edge and horn to punctured-circle open neighborhoods, uses deformation retractions onto those neighborhoods and onto the two-point intersection, and applies open-cover excision. The result is recorded as a Palomar theorem for $\partial\Delta[2]$ only.

7 Acknowledgements

The author expresses sincere gratitude to Professor Dana Scott (Carnegie Mellon University) for suggesting this topic and for inspiring discussions on the formalization of polyhedral geometry. Any remaining errors, misinterpretations, or omissions are solely the author’s own.

7.1 Declaration of Generative AI and AI-Assisted Technologies

In accordance with the Committee on Publication Ethics (COPE) position statements and guidelines on artificial intelligence in research and formal authorship, the author acknowledges the use of large language model tooling during the preparation of this work. Specifically, **Google**

Gemini 3.8, **Cursor Grok 4.6**, and **OpenAI GPT-5.6 Sol** (accessed through **Cursor**) were employed as interactive coding and drafting assistants. They assisted with exploratory formalization, initial Lean 4 proof tactics, the singular-homology and excision investigation, and the structure of the expository drafts. All mathematical claims, Lean 4 proofs, and final text were subsequently reviewed, verified, refined, and certified by the author, who assumes full responsibility for the integrity and accuracy of the content.

8 References

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A Lean module index

Checked by `lake build` against Lean 4 and Mathlib. The development contains no `sorry`, no `admit`, and no project-defined axiom outside `Challenge.lean`. Palomar Comparator compares four theorems: `platonic_solids_3d`, `edges_pos_of_regular`, `regular_polychora_classification`, and `simplicialSingularComparison_boundaryTwo_quasiIso`. Each module below links to its file in <https://github.com/catskillsresearch/platonic-solids>.

A.1 `PlatonicSolids.lean`

Root import of the section-sized modules (41 lines in the repository.)

A.2 [PlatonicSolids/RadialProjection.lean](#)

Frontier of a bounded convex body is homeomorphic to the sphere (25 lines in the repository.)

A.3 [PlatonicSolids/EulerPoincare.lean](#)

Algebraic Euler–Poincaré in dimensions 2 and 3 (112 lines in the repository.)

A.4 [PlatonicSolids/CellularHomology.lean](#)

Cellular Betti finranks and quasi-isomorphism transport (209 lines in the repository.)

A.5 [PlatonicSolids/SingularHomology.lean](#)

Singular $H_0(S^n)$, $H_*(S^0)$, homotopy invariance (238 lines in the repository.)

A.6 [PlatonicSolids/RelativeHomology.lean](#)

Relative singular homology and the LES of a pair (148 lines in the repository.)

A.7 [PlatonicSolids/MayerVietoris.lean](#)

Mayer–Vietoris of an excisive triad; stereographic cover of S^n (303 lines in the repository.)

A.8 [PlatonicSolids/SingularExcision.lean](#)

Root import of barycentric subdivision and small-chain excision (22 lines in the repository.)

A.9 [PlatonicSolids/SingularExcision/Algebra.lean](#)

Ungraded subdivision retract on an abelian group (146 lines in the repository.)

A.10 [PlatonicSolids/SingularExcision/AffineSubdivision.lean](#)

Affine barycentric subdivision and prism (823 lines in the repository.)

A.11 [PlatonicSolids/SingularExcision/AffineRealization.lean](#)

Realization of affine simplices in a space (322 lines in the repository.)

A.12 [PlatonicSolids/SingularExcision/Mesh.lean](#)

Lebesgue number and mesh-ratio $n/(n+1)$ (117 lines in the repository.)

A.13 [PlatonicSolids/SingularExcision/BarycentricGeometry.lean](#)

Diameter of barycentric pieces and eventual smallness (380 lines in the repository.)

A.14 [PlatonicSolids/SingularExcision/SingularChains.lean](#)

Singular Sd as a chain map and the prism homotopy (472 lines in the repository.)

A.15 [PlatonicSolids/SingularExcision/ModuleSupport.lean](#)

Finite coproduct support in ModuleCat (46 lines in the repository.)

A.16 [PlatonicSolids/SingularExcision/Subcomplexes.lean](#)

Lands-in and small singular subcomplexes (174 lines in the repository.)

A.17 [PlatonicSolids/SingularExcision/DegreewisePushout.lean](#)

Degreewise pushout of small chains (109 lines in the repository.)

A.18 [PlatonicSolids/SingularExcision/CokernelPushout.lean](#)

Cokernel model of a preadditive pushout (111 lines in the repository.)

A.19 [PlatonicSolids/SingularExcision/CoverChains.lean](#)

Cover chains isomorphic to small singular chains (133 lines in the repository.)

A.20 [PlatonicSolids/SingularExcision/SmallChains.lean](#)

Small-chain subgroup, invariance, and the retract (1078 lines in the repository.)

A.21 [PlatonicSolids/SingularExcision/OpenCoverQI.lean](#)

Open-cover quasi-isomorphism and `IsExcisiveSubspaces` (523 lines in the repository.)

A.22 [PlatonicSolids/SimplicialSingularComparison.lean](#)

Canonical realization–singular comparison; simplex, horn, and two-vertex cases (3054 lines in the repository.)

A.23 [PlatonicSolids/SimplicialSingularBoundaryTwo.lean](#)

Open-cover gluing proof of the canonical comparison for $\partial\Delta[2]$ (1541 lines in the repository.)

A.24 [PlatonicSolids/SphereSingularHomology.lean](#)

Stereographic $U \cap V \simeq_h S^{n-1}$ and $H_{n+1}(S^{n+1}) \cong H_n(S^n)$ (967 lines in the repository.)

A.25 [PlatonicSolids/Sphere2Homology.lean](#)

Cellular $H_*(S^2; \mathbb{Q})$ of the tetrahedron surface (129 lines in the repository.)

A.26 [PlatonicSolids/Sphere3Homology.lean](#)

Cellular $H_*(S^3; \mathbb{Q})$ of the 4-simplex boundary (195 lines in the repository.)

A.27 [PlatonicSolids/EulerBetti.lean](#)

Betti arithmetic of $\chi(S^2) = 2$ and $\chi(S^3) = 0$ (24 lines in the repository.)

A.28 [PlatonicSolids/PlatonicPair.lean](#)

The five solutions of the Platonic Diophantine inequality (38 lines in the repository.)

A.29 [PlatonicSolids/Reciprocal.lean](#)

$1/p + 1/q > 1/2$ iff the integer inequality (44 lines in the repository.)

A.30 [PlatonicSolids/Incidence.lean](#)

Regularity incidence plus Euler force a Platonic pair and $E > 0$ (84 lines in the repository.)

A.31 [PlatonicSolids/Angles.lean](#)

$\pi/n \in (0, \pi/2)$ so sine and cosine are positive (28 lines in the repository.)

A.32 [PlatonicSolids/GramMatrix.lean](#)

Schläfli Gram matrix and the scalar Δ (30 lines in the repository.)

A.33 [PlatonicSolids/LeadingMinors.lean](#)

Closed forms of the leading minors D_1, D_2, D_3 (34 lines in the repository.)

A.34 [PlatonicSolids/TrailingMinor.lean](#)

Trailing 3×3 minor is the vertex-figure cell minor (22 lines in the repository.)

A.35 [PlatonicSolids/D3IffPlatonic.lean](#)

$D_3 > 0$ iff the cell is Platonic (58 lines in the repository.)

A.36 [PlatonicSolids/QuadForm.lean](#)

Quadratic form of the Gram matrix (26 lines in the repository.)

A.37 [PlatonicSolids/CompletingSquare.lean](#)

LDL / completing-the-square identity (55 lines in the repository.)

A.38 [PlatonicSolids/DetFormula.lean](#)

$\det M = \Delta$ (43 lines in the repository.)

A.39 [PlatonicSolids/Sylvester.lean](#)

Sylvester: positive definite iff leading minors are positive (100 lines in the repository.)

A.40 [PlatonicSolids/TrigValues.lean](#)

Exact
 $\sin^2(\pi/n)$ and
 $\cos^2(\pi/n)$ for n
in

3, 4, 5 (63 lines in the repository.)

A.41 [PlatonicSolids/CompatibleTriples.lean](#)

The eleven Platonic-compatible Schläfli triples (64 lines in the repository.)

A.42 [PlatonicSolids/SignTable.lean](#)

Sign of Δ on those eleven triples (121 lines in the repository.)

A.43 [PlatonicSolids/Regular4.lean](#)

The six regular convex 4-polytopes (28 lines in the repository.)

A.44 [PlatonicSolids/Classification.lean](#)

$\Delta > 0$ (equivalently M positive definite) iff one of the six (84 lines in the repository.)

A.45 [Challenge.lean](#)

Palomar statement of record (deliberate sorries) (158 lines in the repository.)

A.46 [Solution.lean](#)

Palomar solution module: imports PlatonicSolids (25 lines in the repository.)